

HIGH PERFORMANCE COMPUTING FOR SCIENCE AND ENGINEERING

Harvard University
CS 2050

Lecture 14

March 12, 2026



CS-2050 / homework-3

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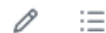


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README



Homework 3

Welcome to Homework 3. This assignment is due by 11:59pm on **Thursday March 12**. Begin with the setup instructions in the next section, and don't forget the submission instructions at the bottom.

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Contributors 2



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Languages



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17 hours ago

README

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About

About

Ask your CS 2050 questions here.

sites.google.com/g.harvard.edu/cs-2050

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31d14b3 · 2 minutes ago

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CS 2050 Final Project

Warning

This is a DRAFT of the final project specification, as of *March 5, 2026*. Please review and ask any clarification questions over the next two weeks. Based on feedback, we will refine and release the final specification at the end of the Spring Recess.

Overview

The final project requires you to apply the high-performance computing techniques learned in class to a computational task of your choice. The project will be completed **individually**. You will begin with a serial version of your algorithm, and then develop parallel versions using OpenMP, MPI, and CUDA. You will analyze its performance in scaling as well as with profiling tools. Finally, you will explore an additional topic beyond the core course material.

Spring Recess

Week 8 - Extended Example: Training Large Language Models



Week 9 - Extended Example: Distributed Linear Algebra



Week 10 - Extended Example: Molecular Dynamics



Week 11 - Extra Topic: Kokkos



Week 12 - Project Work Week



Week 13 - Review



High Performance Computing for Science and Engineering

CS 2050 Challenges

Reminder: Your pseudonym can be found with ``challenge_pseudonym`` on the course cluster.

Last Update: Thursday, Mar 05, 1:51:27 PM

Leaderboards

challenge_0

0 ranked submissions

No results yet.

RULES FOR TODAY

- Group work is encouraged
- No AI tools allowed for the first 10 minutes after each new question release

SETUP

- Request a full node for interactive use

```
srun --nodes=1 --exclusive --time=01:00:00 --pty bash
```

- Load the CS-2050 environment

```
. ~/161588/spack/share/spack/setup-env.sh  
spack env activate -p CS-2050
```


CHALLENGE 6

```
double compute(const pybind11::array_t<double> &X_in,  
               const pybind11::array_t<int> &lengths_in)  
{  
    const int num_rows = X_in.request().shape[0];  
    const int max_columns = X_in.request().shape[1];  
    const double *X = X_in.data();  
    const int *lengths = lengths_in.data();  
  
    double total = 0.0;  
    double total_sq = 0.0;  
    double max_row = std::numeric_limits<double>::lowest();  
  
    #pragma omp parallel for  
    for (int i = 0; i < num_rows; ++i)  
    {  
        double row_sum = 0.0;  
        for (int j = 0; j < lengths[i]; ++j)  
        {  
            const double x = X[i * max_columns + j];  
            row_sum += (std::sin(x) * std::cos(x * x)  
                       + 0.5 * std::sin(0.5 * x + 0.25) *  
std::cos(x + 1.0))  
                        / std::sqrt(j + 1.0);  
        }  
    }
```

CHALLENGE 7

```
#include <pybind11/pybind11.h>
#include <pybind11/numpy.h>
#include <pybind11/stl.h>

#include <cmath>
#include <vector>

#include <omp.h>

// === Header ===
// CHALLENGE_INFO:
// CHALLENGE_ID="challenge7"

// === NOTE ===
// Students should only modify the core of the compute
// function.
// Do not change anything else or your code may not be
// successfully processed.
double compute(const pybind11::array_t<int> &bin_ids_in,
               const pybind11::array_t<double> &weights_in)
{
    const int n = weights_in.size();
    const int *bin_ids = bin_ids_in.data();
    const double *weights = weights_in.data();
```


CHALLENGE 9

```
double compute(const pybind11::array_t<double> &X_in)
{
    const int num_rows = X_in.request().shape[0];
    const int num_columns = X_in.request().shape[1];
    const double *X = X_in.data();

    double result = 0.0;

    #pragma omp parallel for reduction(+ : result)
    for (int i = 0; i < num_rows; ++i)
    {
        for (int j = i; j < num_rows; ++j)
        {
            double dot = 0.0;
            for (int k = 0; k < num_columns; ++k)
            {
                dot += X[i * num_columns + k] * X[j *
num_columns + k];
            }

            const double pair_weight = 1.0 / (1.0 + ((i + j) %
16));
            result += pair_weight * dot + 1.0e-4 * dot * dot;
        }
    }
}
```